

Probability and Distributions

SDS 320E

Basics of Probability

The Role of Probability

Reminder

Statistics can be described as *the art of collecting, organizing, summarizing, and describing data as well as drawing inferences from the data.* - Dr. Amy Maddox

- In statistics, we can never be certain whether our conclusion about our data is right or wrong
- Thus, we try to quantify the uncertainty about our decision using *probability*

Basics of Probability

- Probability is a measure of uncertainty
 - Probability ranges from 0 to 1
 - If $P(A) = 1$, then the event A must always occur
 - If $P(B) = 0$, then the event B must never occur
 - If $P(C)$ is “large” (closer to 1), then that implies that C is very *likely* to occur
 - If $P(C)$ is “small” (closer to 0), then that implies that C is very *unlikely* to occur
-

Basics of Probability Cont.

- Most people confuse *probability* with *possibility*
- *Possibility* is binary such that something either *can* or *cannot* occur
- *Probability* tells us how likely something is to happen, not just *whether* it can happen



What Are the Odds That God Exists? | Young Sheldon

Young Sheldon



Watch on

Note: This is a video on probability vs possibility and not a comment on the existence of a God.

Set Theory

- The mathematical foundation of probability is established through *set theory*
- *Experiment*: a process in which the outcome cannot be determined in advance
 - Ex: rolling a die, flipping a coin, pulling a card out of a deck
- *Sample Space*: set of all possible outcomes of an experiment

Example

Suppose our experiment is flipping a coin 2 times. We want to construct the sample space for the experiment.

Events

➤ Event: a subset of the sample space

Example

Suppose our experiment is flipping a coin 2 times. What is the probability we get two tails?

Different Types of Events

- *Equally Likely Events*: events of a sample space that are equally likely to occur
 - Ex: flipping a coin and getting a head or getting a tail $P(\text{Head}) = 1/2$ and $P(\text{Tail}) = 1/2$
 - *Mutually Exclusive Events*: events that cannot occur at the same time
 - Ex: When we flip a coin, we can only get a head or a tail, but not both
 - *Exhaustive Events*: events that complete or *exhaust* the sample space
 - Ex: Head and Tails are exhaustive because they represent all possible outcomes of flipping a coin
-

Probability Rules

➤ In probability, we often use $\cup$ to denote “or” and $\cap$ to denote “and”

➤ *General Additivity Rule:* $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

➤ Note: If A and B are *mutually exclusive*, then $P(A \cap B) = 0$

$$P(A \cup B) = P(A) + P(B)$$

➤ *Multiplication Rule:* $P(A \cap B) = P(A) \cdot P(B | A) = P(B) \cdot P(A | B)$

➤ *Law of Total Probability:* $P(B) = \sum_i P(A_i) \cdot P(B | A_i)$

➤ *Bayes Theorem:* $P(A | B) = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{P(B|A) \cdot P(A)}{\sum_i P(A_i) \cdot P(B|A_i)}$ by law of total probability

Random Variables

► **Random Variable:** A function that maps the sample space S onto a real numbers

$$X(S) \rightarrow \mathbb{R}$$

► Generally, we denote random variables by X , Y , Z , and ect.

Example

Consider flipping a coin 3 times. Let $X = \text{the numbers of heads}$. Define the random variable X .

Probability Distributions

Probability Distributions

- **Probability distribution:** *describes ALL possible values that can be obtained by a random variable*
 - We denote the probability distribution by $f(X)$
 - The *probability distribution* depends on the type of random variable we have i.e. discrete or continuous
 - If X is discrete, then the probability distribution is given by the *probability mass function (PMF)*
 - If X is continuous, then the probability distribution is given by the *probability density function (PDF)*
-

Probability Mass Function

- **Probability Mass Function (PMF):** The *PMF* of a discrete random variable X is given by

$$f_X(x) = P(X = x).$$

I.e. the *PMF* specifies all possible values of the discrete random variable and their associate probabilities

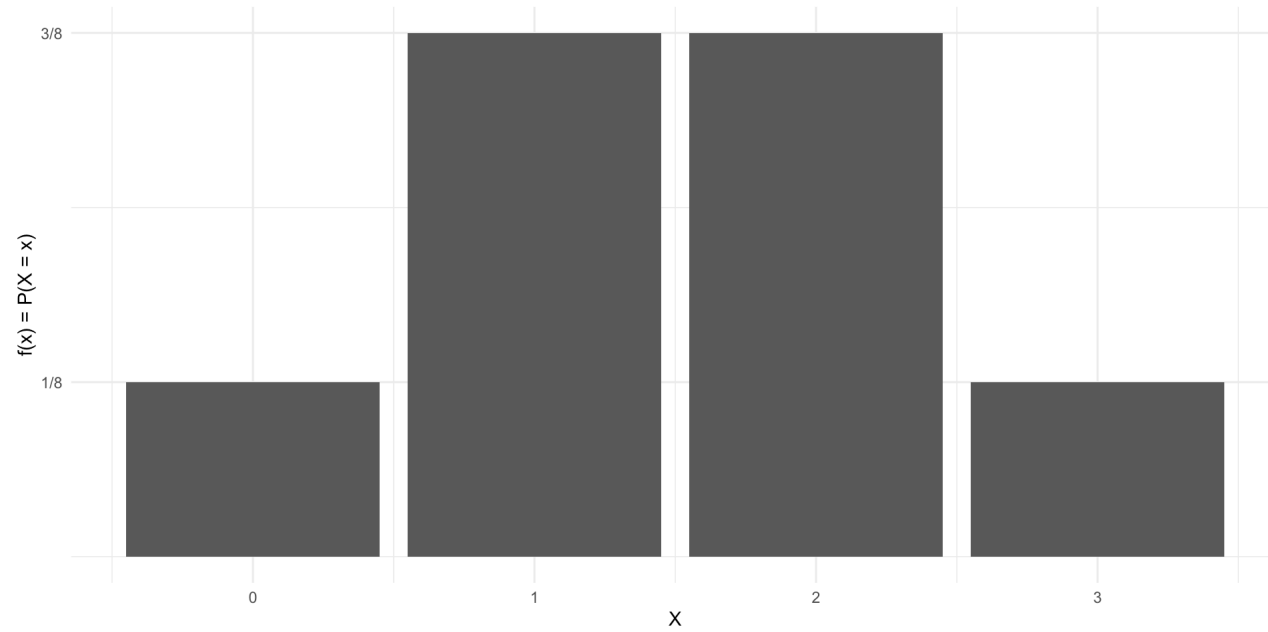
Example

Consider flipping a coin 3 times. Let $X = \text{the numbers of heads}$. Find the *PMF* for X .

We will learn this a special distribution called “Binomial” and the PMF can also be represented by the formula $f(x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x}$, $x = 0, 1, 2, 3$

Visualizing PMF

➤ Bar ➤ Line



PMF Cont.

- We can find specific probabilities using a *PMF*

Example

Consider the PMF below.

X	0	1	2	3	4	5
$f(x)$	0.20	0.10	0.15	0.25	0.18	0.12

Find the following.

1. $P(X \leq 1)$
2. $P(1 \leq X < 3)$
3. $P(X = 7)$

Probability Density Function

Reminder

The probability distribution for *discrete* data is given by the *PMF*, which represents $f_X(x) = P(X = x)$.

- The probability distribution for *continuous* data is given by the *probability density function (PDF)*, which represents ...

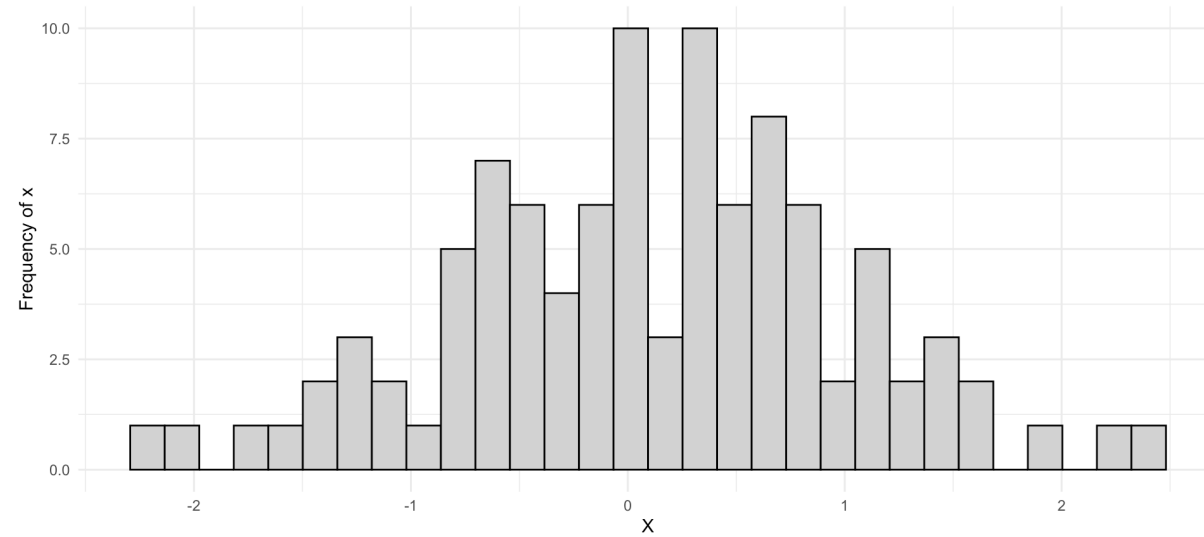
Issue

For continuous data, we generally do not have probability about a specific value. If X is continuous, then for a random value x , $P(X = x)$

- Thus, the *PDF* does not represent probability for continuous data, rather it represents the “density” of observations i.e. frequency of occurrence

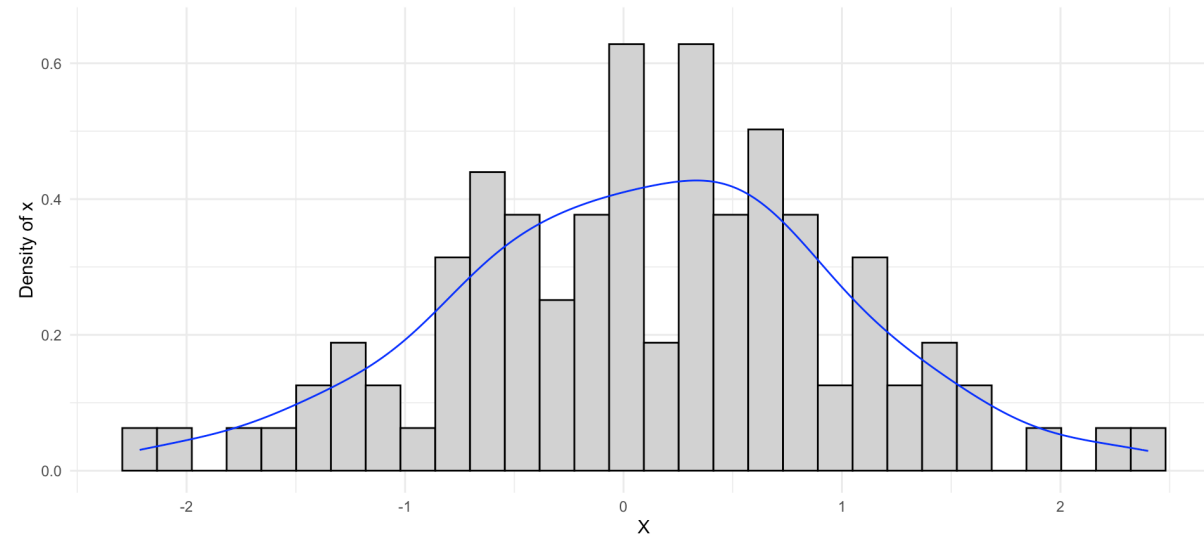
Visualizing PDF

► We have seen frequency for continuous data before ...



Visualizing PDF Cont.

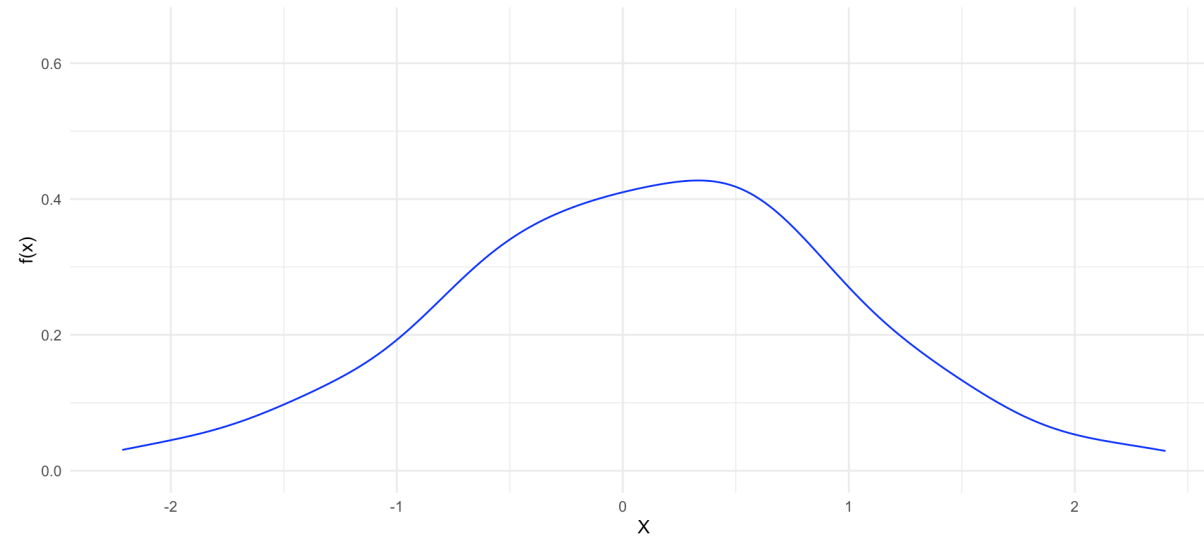
- Rather than using frequency, we could use the “density”



- Lastly, we could represent the histogram with a “smoothed out curve”

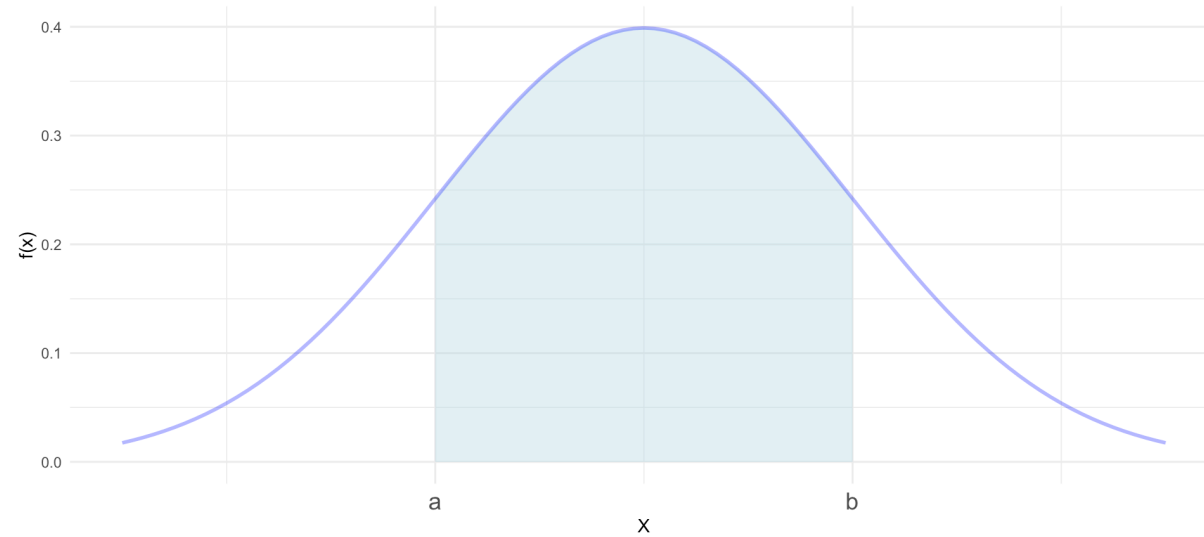
Visualizing PDF Cont.

- The *probability density function*, $f(x)$, is the resulting “smoothed out curve”



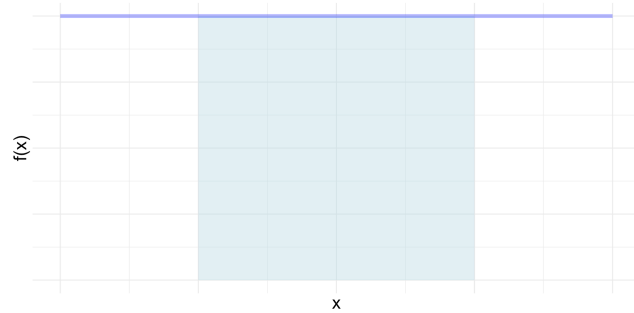
Probability with PDF

➤ To find $P(a \leq X \leq b)$ with a *PDF*, we find the area under *PDF* curve from a to b

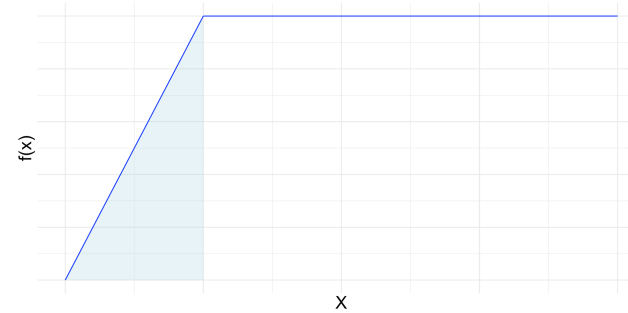


Probability with PDF Cont.

- When the shape of the $f(x)$ is “geometrically pleasing”, finding area under the *PDF* curve is straight forward



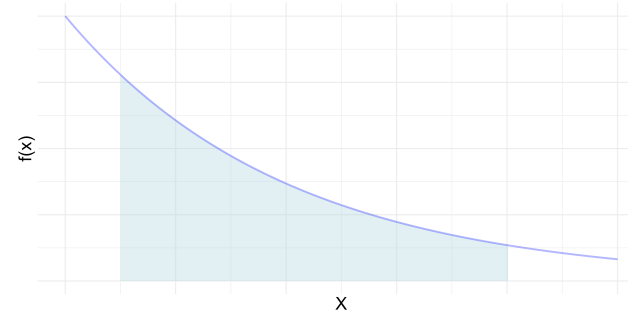
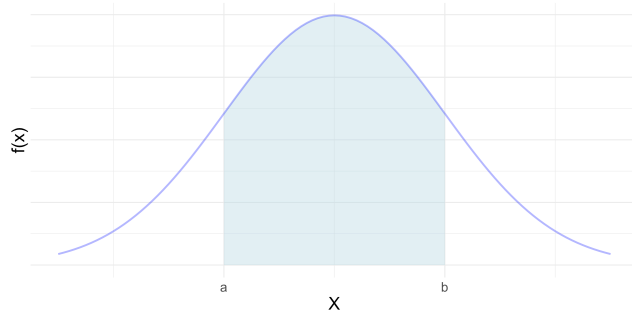
Area = bh



Area = $1/2bh$

Probability with PDF Cont.

➤ For other *PDFs*, finding probability requires calculus $\int_a^b f(x)dx$



➤ This calculus is represented by the *cumulative distribution function* ...

Cumulative Distribution Function

- **Cumulative Distribution Function (CDF):** *the CDF of a random variable gives cumulative probability up to a specific value and is denoted by*

$$F_X(x) = P(X \leq x)$$

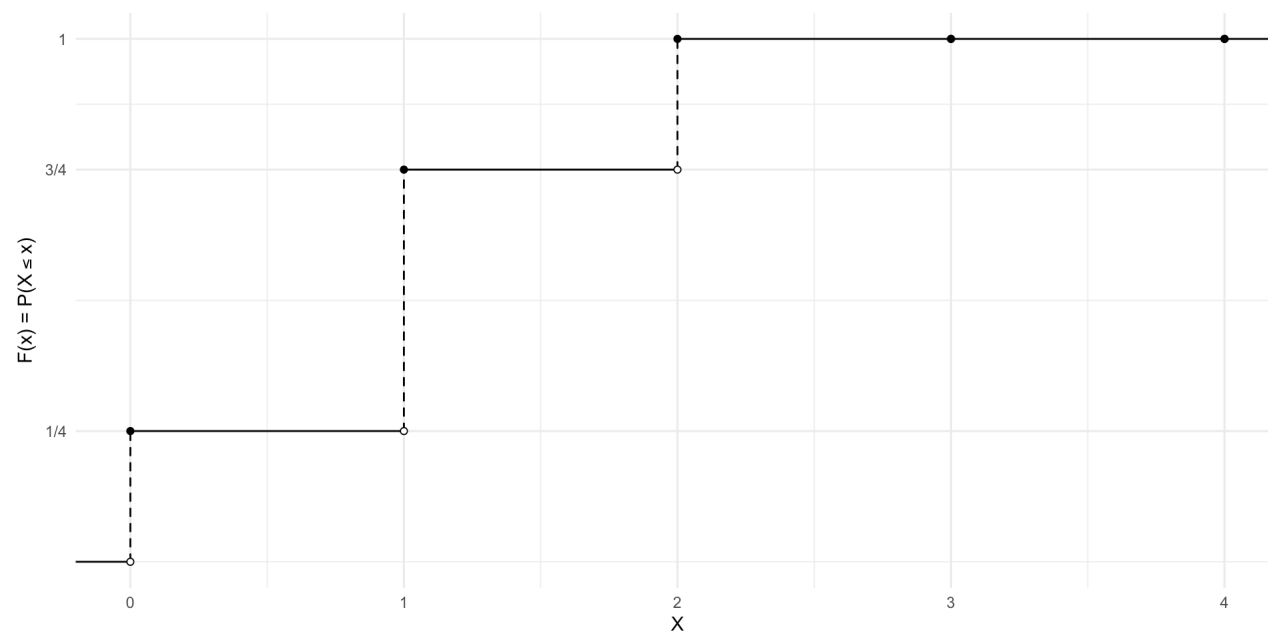
Example

Consider flipping a coin twice. We let $X = \text{the number of heads}$. We can easily show that the *PMF* of X is given by below.

X	0	1	2
$f(x)$	1/4	2/4	1/4

Find the *CDF* of X .

Visualizing CDF



Probability with CDF

- We can find specific probabilities using a *CDF*. The use of the *CDF* will follow these general formulas

$$\text{► } P(X \leq a) =$$

$$\text{► } P(X \geq a) =$$

$$\text{► } P(a \leq X \leq b) =$$

Warning

Note: The *CDF* evaluations, $F(\quad)$, are intentionally left blank. You must determine the value to evaluate based on:

1. The type of data (discrete or continuous)
2. The possible values of the random variable

Probability with CDF Cont.

Example

Consider the CDF below.

X	-1	2	3	5	6	7
$F(x)$	0.20	0.30	0.45	0.70	0.88	1

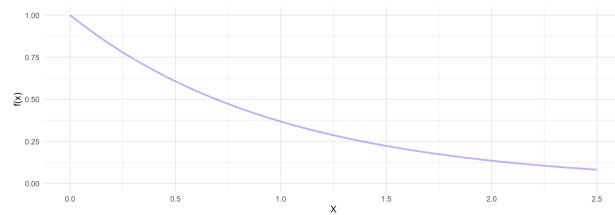
Find the following.

1. $P(2 < X \leq 5)$
2. $P(X \geq 2)$
3. $P(2 \leq X < 5)$

Probability with CDF Cont.

Example

Consider a continuous random variable X .



$$\text{calculus} \rightarrow F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & x \geq 0 \end{cases}$$

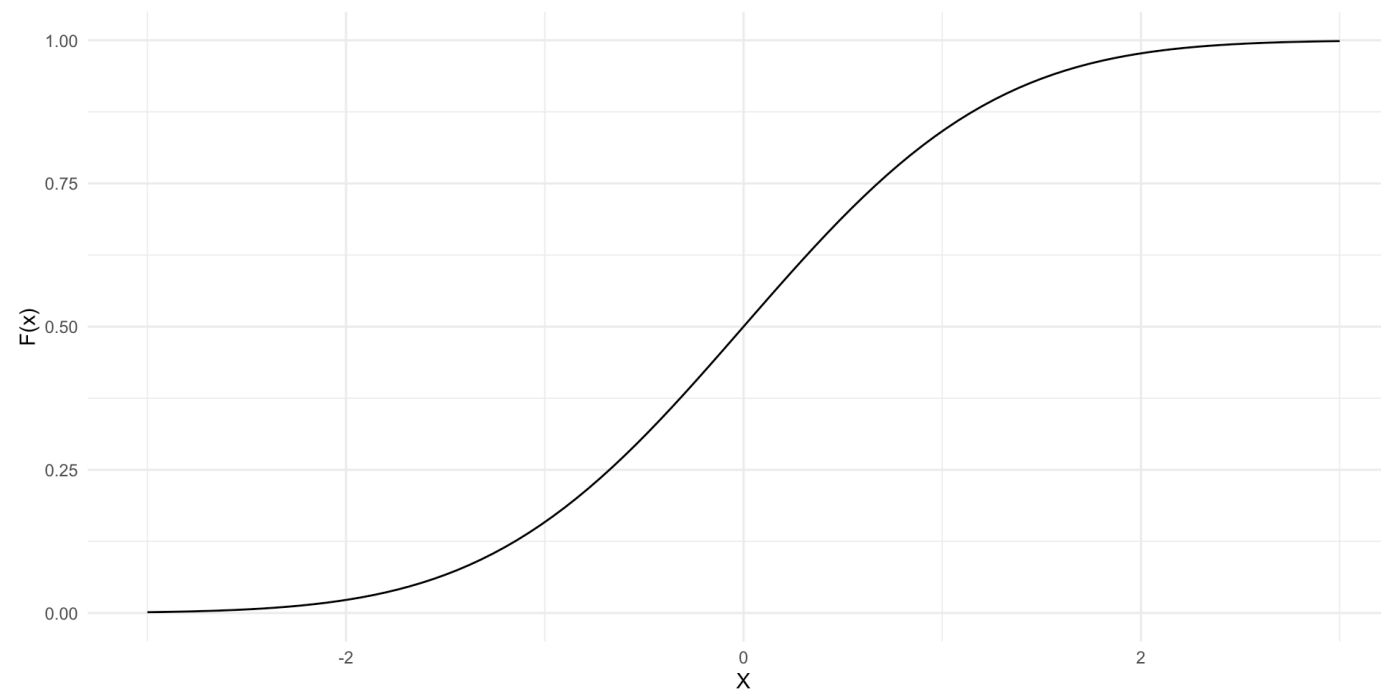
Find $P(2 \leq X \leq 5)$.

$= F(5) - F(2)$ note: since X is continuous, there is no mathematical difference between $\leq$ or $<$

$$= 1 - e^{-5} - (1 - e^{-2})$$

$$= 0.1286$$

Visualizing Continuous CDF

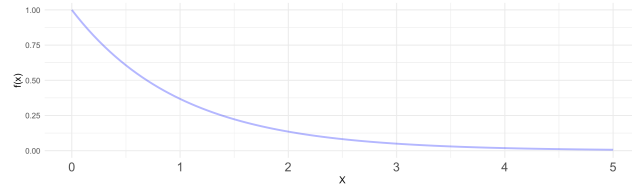
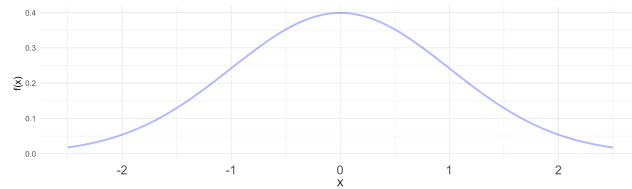


Special Distributions

i Note

Certain “problems” or random variables will follow specific distributions that are well known and have closed form representations of their probability distributions.

X	0	1	2	3
$f(x)$	1/8	3/8	3/8	1/8



$$\rightarrow f(x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x}, x = 0, 1, 2, 3$$

$$\rightarrow f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}, -\infty < X < \infty$$

$$\rightarrow f(x) = e^{-x}, 0 \leq X < \infty$$

Binomial Distribution

Binomial Distribution Definition

Binomial Distribution

A discrete random variable $X = \text{the number of successes}$, with n trials and probability of success p , is said to follow a *binomial distribution*, denoted

$X \sim \text{Bin}(n, p)$, note: $\sim$ denotes “distributed”

if it satisfies the following:

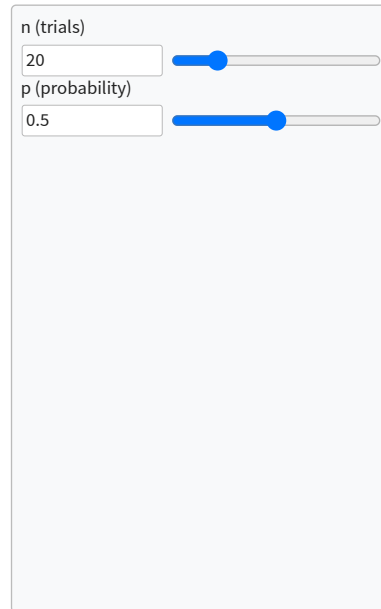
1. $X = \text{the number of successes}$
2. There are only two outcomes, with constant probability of success p and probability of failure $1 - p$
3. There are a finite number of n independent trials

➤ Binomial Distribution Examples:

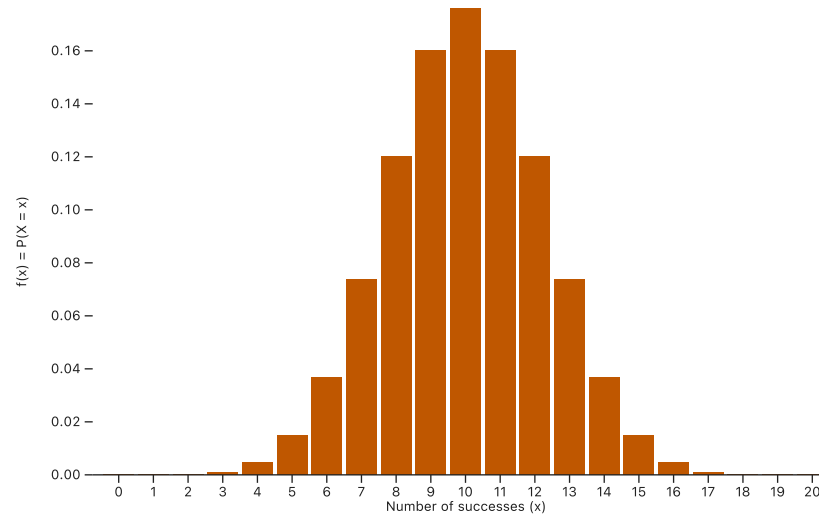
- Flipping coin 2 times and $X = \text{number of heads}$. Thus, $X \sim \text{Bin}(n = 2, p = 0.5)$.
- Rolling die 4 times and $X = \text{number of 5s}$. Thus, $X \sim \text{Bin}(n = 4, p = 1/6)$.
- Manufacturing process and $X = \text{number of defects}$ and there is constant probability of defect occurring 5% of the time. Thus, $X \sim \text{Bin}(n, p = 0.05)$.

Visualizing Binomial Distribution

- Adjust the number of trials (n) and probability of success (p) to see how the distribution changes.



Binomial($n = 20, p = 0.50$) Distribution



PMF for Binomial

i Binomial PMF

Let $X \sim \text{Binomial}(n, p)$. Then, the *probability mass function* for X is given by

$$f_X(x) = \binom{n}{x} (p)^x (1-p)^{n-x}, \text{ for } x = 0, \dots, n.$$

note: $\binom{n}{x} = \frac{n!}{x!(n-x)!}$

➤ To get the *PMF* of binomial distribution in R, we can use the `dbinom()` function. It has the following arguments:

- `x`: the number of successes to find the probability of
- `size`: the number of trials n
- `prob`: the constant probability of success p

```
1 # Suppose X ~ Binomial(n = 10, p = 0.30) and we want P(X = 8)
2 dbinom(x = 8, size = 10, prob = 0.30)
```

```
[1] 0.001446701
```

CDF for Binomial

- The *cumulative distribution function* of a *binomial* random variable is represented by the sum of the *PMF* and is given by

$$F_X(x) = P(X \leq x) = \sum_{k=0}^x \binom{n}{k} (p)^k (1-p)^{n-k}$$

- To get the *CDF* of binomial distribution in R, we can use the `pbinom()` function

```
1 # Suppose X ~ Binomial(n = 10, p = 0.30) and we want P(X <= 3)
2 pbinom(3, size = 10, prob = 0.30)
```

```
[1] 0.6496107
```

Binomial Example

Example

Consider flipping a coin 20 times. Let X = the number of heads. Find the PMF of X .

Before, we would first find the sample space ...

$S = \{hhhhhhhhhhhhhhhhhhhh, tttttttttttttttt, httttttttttttttt, \dots$

There are $2^{20} = 1,048,576$ possible outcomes in the sample space 😞

instead...

- We have a finite number of
- Only two outcomes with

Binomial Example Cont.

Example

Consider flipping a coin 20 times. Let X = the number of heads. Find the following:

1. $P(X = 10)$

2. $P(8 \leq X < 12)$

```
1 pbinom( , 20, 0.50) - pbinom( , 20, 0.50)
```

```
[1] 0.6166897
```

Normal Distribution

Normal Distribution Definition

Normal Distribution

Suppose X is a continuous random variable with a mean μ and standard deviation σ . Further suppose that X is normally distributed, denoted as

Then, the *probability density function* (PDF) of X is given by

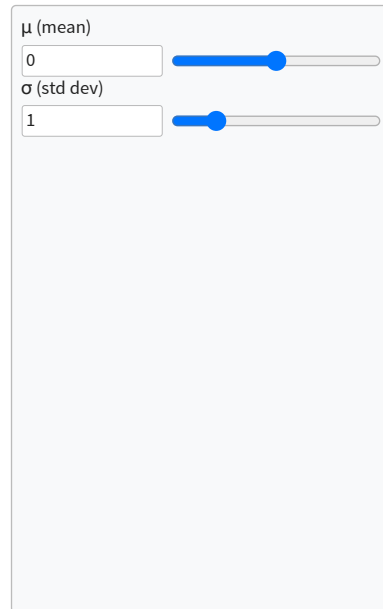
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}, \text{ for } -\infty < X < \infty$$

➤ Normal Distribution Examples:

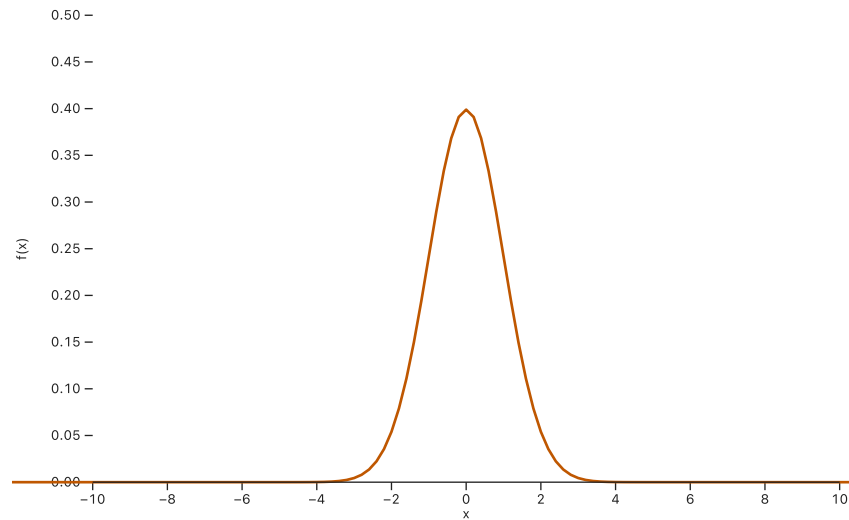
- A lot of data is “normally” distributed! Hence, the name “Normal” distribution the original name is gaussian distribution
 - Human height (within a sex)
 - Weight
 - Stock data
 - Body temperature
 - and many more ...

Visualizing Normal Distribution

- Adjust the mean μ and standard deviation σ to see how the normal distribution changes



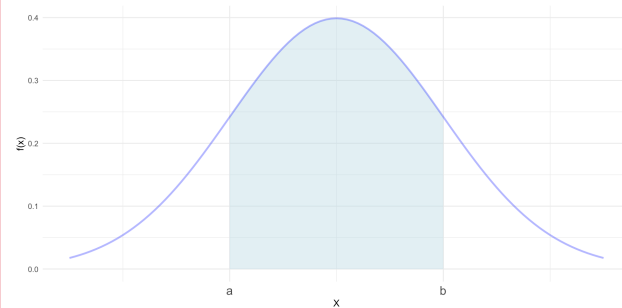
Normal($\mu = 0.0$, $\sigma = 1.0$) Distribution



Probability with Normal Distribution Issue

! Reminder

To find probability with a Normal *PDF* curve, it requires calculus



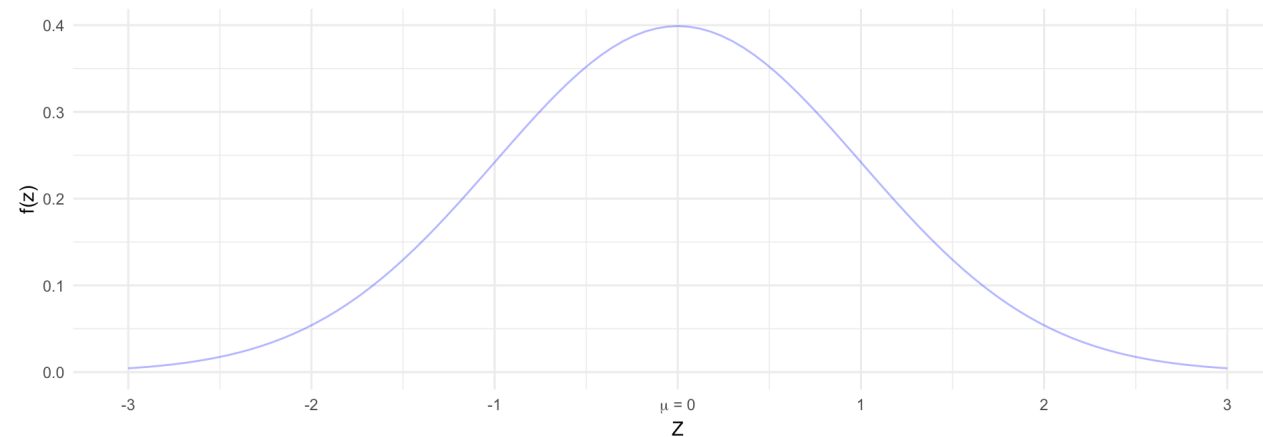
$$\rightarrow \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} dx$$

- To find probability for Normal distribution by hand, we will use the *Standard Normal Distribution*

Standard Normal Distribution

Standard Normal Distribution

The *Standard Normal* distribution is a special case of the *Normal* distribution with mean $\mu = 0$ and $\sigma = 1$ and is denoted by



- The *cumulative distribution function* (CDF) of Z has been evaluated and written in tables: [Z-Table link](#)

Standardizing Normal Distribution

Z-Score

Each *Normal* distribution can be standardized using the formula

$$Z = \frac{X - \mu}{\sigma}$$

z is referred to as the *z-score*.

- The *z-score* represents how many *standard deviations* the value X is above or below the mean

Example

Suppose $X \sim \mathcal{N}(20, 5)$. Find the *z-score* for the following.

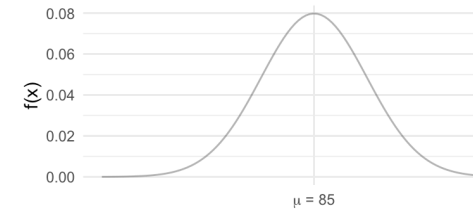
1. $X = 25$
2. $X = 10$

Probability with Normal Distribution by Hand

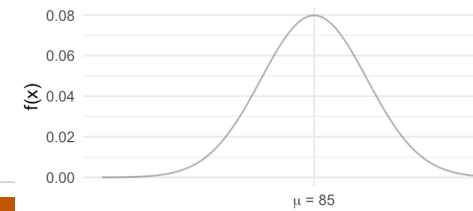
Example

Let $X = \text{Exam 1 scores for previous SDS 320E Semesters}$, and it is known the the scores are normally distributed with a mean of 85 and standard deviation of 5. Find the following.

1. $P(75 \leq X < 90)$



2. The Median Score was an 89.5. What is the Probability a random student scored higher than the Median?

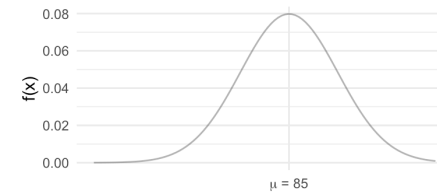


Quantiles with Normal Distribution

Example

Let $X = \text{Exam 1 scores for previous SDS 320E Semesters}$, and it is known the the scores are normally distributed with a mean of 85 and standard deviation of 5.

Suppose a student scored in the 97.5th percentile. What score did they get?



Normal Distribution in R

- To get the CDF of a normal distribution in R, we can use the `pnorm()` function. It has the following arguments:
 - `q`: the numerical value to evaluate
 - `mean`: the mean μ
 - `sd`: the standard deviation σ
 - `lower.tail`: logical; if TRUE (default), probabilities are $P(X \leq x)$, otherwise, $P(X > x)$.

```
1 # X ~ N(27, 2.5) and we want P(X < 23)
2 pnorm(23, mean = 27, sd = 2.5)
```

```
[1] 0.05479929
```

- We can also get quantiles using the `qnorm()` function

```
1 # X ~ N(27, 2.5) and 90th percentile
2 qnorm(0.90, mean = 27, sd = 2.5)
```

```
[1] 30.20388
```

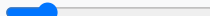
Sampling Distribution and CLT

Statistic Review

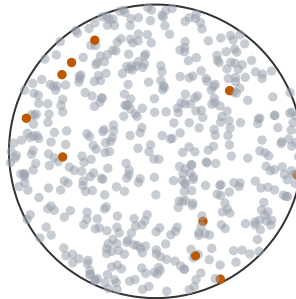
Reminder

A **Statistic** is a measure of the sample.

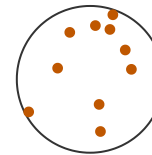
Sample size (n)

10 

Population



Sample



obs value

1	61.08
2	39.31
3	61.85
4	73.24
5	43.73

$$\bar{x} = 51.789$$

➤ Thus, each time we take a sample, the statistic will vary

Sampling Distribution

➤ Since a statistic will vary, it will have a probability distribution

Statistic
Mean ▼

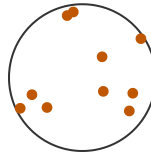
Sample size (n)
10

Draw Sample

Number of samples
100

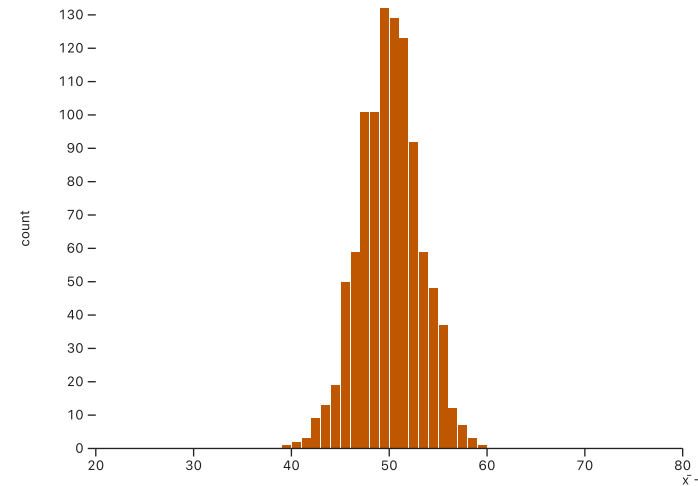
Draw Many Samples

Sample



$\bar{x} =$
49.325

Distribution of $\bar{x}$ (n = 10, 1001 samples drawn)



➤ The probability distribution of a statistic is referred to as the **sampling distribution**

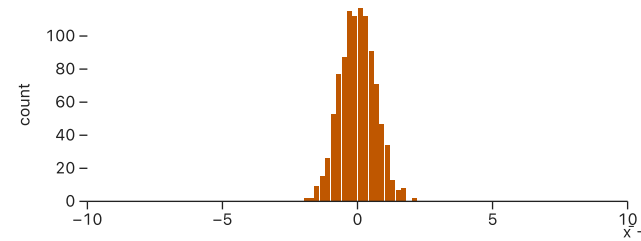
Distribution of $\bar{X}$ when X is Normal

Example

Suppose $X \sim \mathcal{N}(\mu, \sigma)$. What is the distribution of $\bar{X}$?



$\mu = 0.0, \sigma = 2.0, n = 10, 1000 \text{ samples}$



Mean of $\bar{x} = 0.005$

SD of $\bar{x} = 0.648$

- $\bar{X}$ looks normally distributed!
 - It has the same mean as μ
 - But a slight different standard deviation?

Distribution of $\bar{X}$ when X is Normal Cont.

Theorem

Suppose X is normally distributed with mean μ and standard deviation σ , i.e $X \sim \mathcal{N}(\mu, \sigma)$. Then, the sampling distribution of $\bar{X}$ is

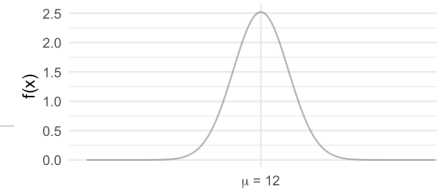
- Note: This result holds true regardless of the sample size! Sample size will matter for CLT
- Since we know the distribution of $\bar{X}$, we can do the following:
 - Find probability about $\bar{X}$
 - Make inference about $\bar{X}$

Probability of $\bar{X}$ when X is Normal

Example

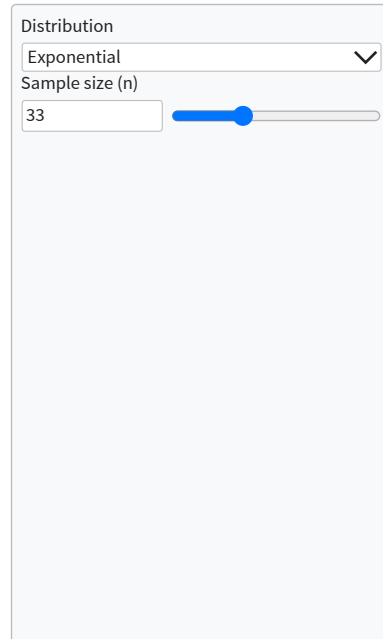
Coca-Cola cans are labeled as 12 oz. However, the 12 oz is not an exact amount in every single can. It may vary slightly from can to can. Suppose the fill amount of Coca-Cola cans is normally distributed with mean of 12 oz and standard deviation of 0.5 oz. Find the following.

1. What is the distribution of $\bar{X}$ = the fill amount of Coca-Cola cans?
2. Suppose we take a sample of 10 Coca-Cola cans, what is the distribution of the mean fill amount?
3. What is the probability the average fill amount for a sample of 10 is less than 11.8oz?

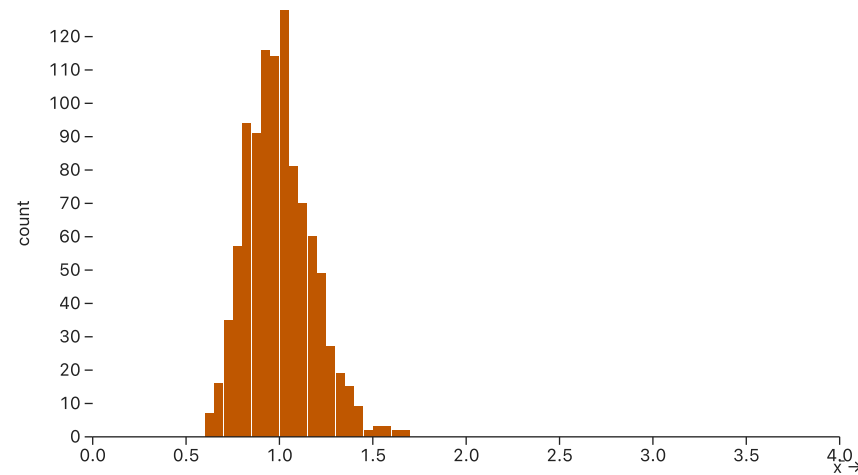


Central Limit Theorem

► What is the distribution of $\bar{X}$, if X is not normal?



Distribution of $\bar{x}$ for Exponential, $n = 33$, 1000 samples



Central Limit Theorem Cont.

Central Limit Theorem (CLT)

Suppose X is distributed with mean μ and standard deviation σ . Then, for a “sufficiently large” sample size n , the sampling distribution of $\bar{X}$ is

note: $\sim$ means approximately distributed

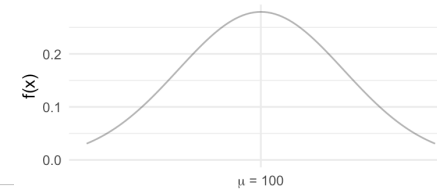
- “Sufficiently large” sample size will depend on the underlying distribution i.e. more skewed data will need larger n to be sufficient
 - A crude rule for sufficiently large is if $n > 30$
- The CLT is one of the most important/popular theorems in statistics
 - It allows us to make inference about a mean with very little information

Probability for CLT

Example

Medical labs periodically check that their testing equipment is reading accurately by running a control solution with a known, certified concentration. Suppose this control solution has a glucose concentration with a mean of 100 mg/dL and a standard deviation of 10 mg/dL. Find the following

1. What is the distribution of X = glucose concentration?
2. Suppose a lab want to check 49 of their testing equipment. What is the distribution of the average glucose concentration for a sample of 49?
3. What is the probability that for a sample of 49 testing equipment, the average glucose concentration is greater than or equal to 102?



The End

